# Siddharth Deshpande

ARCHITECT | APPLIED AI | PRODUCTION SYSTEMS

+91 78478 96052 · dsiddharth2@gmail.com
siddharthdeshpande.com · github.com/dsiddharth2
linkedin.com/in/dsiddharth2

**12 years** turning ambiguous business problems into production systems – ERPs, AI platforms, multi-agent pipelines – across startups and enterprise. I've been the direct client interface since 2016 and run **3 client engagements** today; I don't just write code, **I own the outcome end to end.** At **Apra Labs** I built the AI and data platform layer serving **300+ US facilities**: **7 AI agents** and **5 production AI systems**, a GraphRAG engine over **20K support documents**, conversational AI across a **4 TB database**, and a FinOps agent that cut cloud spend **30%** – while hiring and mentoring **20+ engineers** and growing with the company **from 10 to 55**. Before that, a logistics ERP built from scratch to **2,500 daily users**, **70K dockets/month** and **100K hits a day**, at **MildlyClassic**, where our team also built the software that got **Wifi Dabba into YC W17**. Separately, I co-founded **UXgage**, a behavioral analytics startup **₹5L funded by Deshpande Startups, Hubli**.

## EXPERIENCE

### Apra Labs

**Architect · previously Associate Architect**

Aug 2019 – Present

Built the **AI and data platform layer** for a unified facility management product serving **300+ US facilities**. Managing **3 clients**, mentoring a **team of 4**, **architected all systems end-to-end**. **Campus hiring lead for 4 years**; grew with the company **from 10 to 55**.

### MildlyClassic

**Engineering Team Lead**

May 2016 – May 2019

Built and scaled a **logistics ERP from scratch** to **2,500 daily users** handling **70K dockets/month** for PAN India secure jewellery logistics. **Hired 7 engineers**, established a **4-week release cycle**, migrated **DigitalOcean → AWS in 3 months**.

*Sister company · **Wifi Dabba → YC W17**:* built **byte-counting and usage sync software** on Wi-Fi hardware, and the **Google NBU analytics platform**.

### UXgage

**Co-Founder & CTO**

Oct 2015 – Apr 2016

Took an idea **from zero to a shipped product in 6 months** — won a **₹5L cash prize** from Deshpande Startups, Hubli, with office space and Azure infrastructure granted separately.

### Earlier

2010 – 2016

**Samvit Learning**, Founder — software training institute founded during MCA; trained **250+ students**. **Firaa.me**, side project at IdeaFlask — travel blogging platform, **500+ organic users**. **dotCORD IT Solutions**, Intern — **3 years** at GIT Incubation Center.

## OPEN SOURCE

**Apra Agent Kit** — starting point for a new agent. **/agent-builder** in Claude writes the spec, the plan, and the build. **12 tools**.

**Plug** — package manager for Claude Code skills; install and manage reusable Skills, Commands and Agents from GitHub-hosted registries. **14 packages**, on npm as *plugvault*.

**CodeHawk** — AI pull request review pipeline. Runs in CI, scores code quality and posts inline comments. **16 language rule sets**, **394 unit tests**.

**PHP ZXing** — PHP wrapper for the ZXing barcode library. **277K+ Packagist installs**, **133 GitHub stars**, featured on the upstream ZXing repo.

**MooPin** — Pinterest-style column grid for MooTools, incl. infinite scroll. **~120 lines**.

## TECHNICAL EXPERTISE

**Languages & Frameworks** — Python · TypeScript · C# / .NET 8 · PHP · React · Node.js · LangChain · LangGraph

**AI & Data** — GPT / LLM integration · FAISS · Vector search · GraphRAG · Multi-agent systems

**Infrastructure** — Azure (Durable Functions, Service Bus, Cosmos DB) · AWS · Neo4j · Cassandra

**Practices** — System design · CI/CD · DevOps · FinOps · Hiring & mentorship

## EDUCATION & RECOGNITION

**Master of Computer Applications (MCA)** — Gogte Institute of Technology, Belgaum

**Leadership Excellence**, Apra Labs (Nov 2024) · **Customer Appreciation**, Apra Labs (Apr 2023) · **Guiding Vision**, Pravega X Fest

## SELECTED CASE STUDIES

### GraphRAG Knowledge Engine

*Enterprise knowledge graph pipeline*

Python · Azure Durable Functions · Cosmos DB · FAISS · gpt-4o-mini

Builds **knowledge graphs** with **entities, relationships, topics and communities** via **vector similarity**

**Two-stage ingest**: **seconds-to-searchable indexing**, then **graph enrichment** in the background

**90K entities** and **150K+ relationships** across **11 Cosmos collections**

**Outcome** — **Support knowledge reusable at scale**; duplicate ticket resolution dramatically reduced

### Multi-Agent Conversational AI

*Supervisor-pattern agent orchestration*

Python · LangChain · LangGraph · Azure SQL · Azure SignalR · React

Supervisor pattern routing to **5 specialized agents** across **~30 registry-driven tools**

Natural-language SQL with **permission rewrite**, seven-collection RAG, **100+ report types** over a **4 TB** database, SignalR streaming of the reasoning trace

**Outcome** — Report generation reduced from **days to seconds**; non-technical staff querying data independently

### Cloud Cost Intelligence Agent

*Automated FinOps analysis*

TypeScript · Azure Cost Management · Resource Graph · Claude Sonnet

**Nightly FinOps agent**: **19 collectors** across 3 phases, **13 anomaly detectors**, 4-tier breach thresholds

**20+ standing questions** answered deterministically; the LLM handles only reasoning, with a **doer–reviewer loop** checking claims against raw files

**Outcome** — **30% cloud cost reduction**; prioritized recommendations delivered nightly, teams now own their spend

### Infrastructure Performance Agent

*Health scoring & regression detection*

TypeScript · Azure Monitor · SQL DMVs · Query Store · Service Bus

Composite **0–100 health score** per service, with **rolling 30-day baselines** for **regression detection**

Azure Monitor metrics and SQL DMV telemetry sampled at **15-second intervals**, **13 anomaly checks**, 4-tier breach thresholds

**Outcome** — Degradation surfaced **before user impact**; **alert fatigue significantly reduced**

### Secure Logistics ERP

*PAN India logistics platform*

PHP · AWS · React · Node.js · MySQL

**End-to-end ERP built from scratch — Finance, Operations, Booking, Hub Management, CRM, Route Planning**

**2,500 daily users**, **70K dockets/month**, **100K daily hits**, zero downtime through scaling

**Outcome** — **End-to-end shipment visibility** for all clients; **manual processes reduced by over 80%**

### AI Code Review Pipeline

*CI/CD-integrated automated reviews*

Python · Azure DevOps · GPT · REST APIs

Reviews pull requests with **inline comments and star ratings**. **10+ languages**, **penalty-based scoring**, **fix verification mode**

**Outcome** — **Review bottleneck for senior engineers eliminated**; consistent scoring across all repos

### Search Index Sync Engine

*Real-time & bulk search indexing*

C# / .NET 8 · Azure AI Search · Service Bus · Durable Functions

**Bulk indexing** via Durable Functions and **real-time incremental sync** via Service Bus, with **index versioning** and a **dead letter queue**

**Outcome** — **Entity search across 7 people-data entity types** made near-instant; search freshness **under 5 seconds**, zero-downtime schema migrations

### UXgage – Behavioral Analytics

*Enterprise usability platform*

PHP · Neo4j · Cassandra · JavaScript

Captures clicks, mouse movements, text input and URL transitions to **reconstruct actual user workflows**. **Built the entire stack from scratch**

**Outcome** — **₹5L seed funding**; idea to funded product in under 6 months